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STAT2920: Introduction to Probability

Tutorial for Chapter 5: Special Probability Distributions

Exercise 1.
Five fair coins are flipped. If the outcomes are assumed independent,

(a) find the probability mass function of the number of heads obtained;

(b) find the mean and variance of the number of heads obtained.

Solution 1.
(a) Let X be the number of heads that appear, then X is a binomial random variable with parameters $(n = 5, \theta = 1/2)$. Hence,

$$\begin{aligned}
P(X=0) &= \binom{5}{0}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^5 = \frac{1}{32},\\
P(X=1) &= \binom{5}{1}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^4 = \frac{5}{32},\\
P(X=2) &= \binom{5}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 = \frac{10}{32},\\
P(X=3) &= \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = \frac{10}{32},\\
P(X=4) &= \binom{5}{4}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^1 = \frac{5}{32},\\
P(X=5) &= \binom{5}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0 = \frac{1}{32}.
\end{aligned}$$

(b) Since $X \sim \mathsf{Bin}(5, 1/2)$,

$$E[X] = 5 \cdot \frac{1}{2} = \frac{5}{2}, \quad Var(X) = 5 \cdot \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{5}{4}.$$

□

Exercise 2.
An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn,

(a) what is the probability that exactly n draws are needed;

(b) compute the mean and variance of the number of draws needed.

Solution 2.
Let X denote the number of draws needed to select a black ball, then $X \sim \mathsf{Geo}(\theta = M/(M+N))$.

(a)

$$P(X=n) = \frac{M}{M+N}\left(\frac{N}{N+M}\right)^{n-1} = \frac{MN^{n-1}}{(M+N)^n}.$$

(b) One has

$$E[X] = \frac{1}{\frac{M}{M+N}} = \frac{M+N}{M}, \quad Var(X) = \frac{1-\frac{M}{M+N}}{\left(\frac{M}{M+N}\right)^2} = \frac{N(M+N)}{M^2}.$$

□

Exercise 3.
A company establishes a fund of 120 from which it wants to pay an amount, C, to any of its 20 employees who achieve a high performance level during the coming year. Each employee has a 2% chance of achieving a high performance level during the coming year. The events of different employees achieving a high performance level during the coming year are mutually independent.

Calculate the maximum value of C for which the probability is less than 1% that the fund will be inadequate to cover all payments for high performance.

Solution 3.
Let X denote the number of employees who achieve the high performance level. Then X follows a binomial distribution with parameters $n = 20$ and $\theta = 0.02$. Now we want to determine x such that $\mathbb{P}[X > x] < 0.01$ or equivalently

$$0.99 \leq \mathbb{P}[X \leq x] = \sum_{k=0}^{x} \binom{20}{x} (0.02)^k (0.98)^{20-k}$$

The first three probabilities (at $0, 1,$ and 2) are $0.668, 0.272,$ and 0.053. The total is 0.993 and so the smallest x that has the probability exceed 0.99 is 2. Thus $C = 120/2 = 60$. □

Exercise 4.

(a) Find the probability that exactly 7 times one must throw a fair die until the outcome 1 has occurred 4 times.

(b) Find the expected value and the variance of the number of times one must throw a fair die until the outcome 1 has occurred 4 times.

Solution 4.
Let X be the number of throwing to obtain outcome 1 exactly 4 times, thus $X \sim \mathsf{NB}(4, 1/6)$.

(a)

$$P(X=7) = \binom{7-1}{4-1}\left(\frac{1}{6}\right)^{4-1}\left(1-\frac{1}{6}\right)^{7-4}\frac{1}{6} = 0.0089.$$

(b) One has

$$E[X] = \frac{4}{1/6} = 24, \quad Var(X) = \frac{4(1-1/6)}{(1/6)^2} = 120.$$

□

Exercise 5.
If the probabilities of having a male or female child are both 0.5, find the probabilities that

(a) a family's fourth child is their first son;

(b) a family's seventh child is their second daughter.

Solution 5.
(a) This is a probability of geometric distribution, $x = 4, \theta = 1/2$.

$$P(X = 4) = \frac{1}{2} \cdot \left(1 - \frac{1}{2}\right)^{4-1} = \frac{1}{16}.$$

(b) This is a probability of negative binomial distribution, $x = 7, \theta = 1/2, r = 2$.

$$P(X = 7) = \binom{7-1}{2-1} \left(\frac{1}{2}\right)^{2-1} \left(1 - \frac{1}{2}\right)^{7-2} \frac{1}{2} = 0.0469.$$

□

Exercise 6.
A group of 100 patients is tested, one patient at a time, for three risk factors for a certain disease until either all patients have been tested or a patient tests positive for more than one of these three risk factors. For each risk factor, a patient tests positive with probability p, where $0 < p < 1$. The outcomes of the tests across all patients and all risk factors are mutually independent.

Determine an expression for the probability that exactly n patients are tested, where n is a positive integer less than 100.

Solution 6.
From the binomial distribution formula, the probability P that a given patient tests positive for at least 2 of these 3 risk factors is

$$P = \binom{3}{2} p^2 (1-p)^{3-2} + \binom{3}{3} p^3 (1-p)^{3-3} = 3p^2(1-p) + p^3.$$

Using the geometric distribution formula with probability of success $P = 3p^2(1-p) + p^3$, the probability that exactly n patients are tested is

$$(1-P)^{n-1} P = \left[1 - 3p^2(1-p) - p^3\right]^{n-1} \left[3p^2(1-p) + p^3\right]$$

□

Exercise 7.
A box contains 10 red balls and 15 green balls. If 5 balls are drawn at random without replacement, what is the probability that exactly 3 of the drawn balls are red?

Solution 7.
The problem involves a hypergeometric distribution since we are drawing balls without replacement. The probability mass function (PMF) for the hypergeometric distribution is given by:

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

Where:

- $N = 25$ is the total number of balls (10 red + 15 green),
- $M = 10$ is the number of red balls,
- $n = 5$ is the number of balls drawn,
- $x = 3$ is the number of red balls drawn.

Now, we compute the values:

$$P(X = 3) = \frac{\binom{10}{3}\binom{15}{2}}{\binom{25}{5}}$$

First, calculate the binomial coefficients:

$$\binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$

$$\binom{15}{2} = \frac{15 \times 14}{2 \times 1} = 105$$

$$\binom{25}{5} = \frac{25 \times 24 \times 23 \times 22 \times 21}{5 \times 4 \times 3 \times 2 \times 1} = 53130$$

Thus, the probability is:

$$P(X = 3) = \frac{120 \times 105}{53130} \approx 0.238$$

The probability that exactly 3 red balls are drawn is approximately 0.238.

Exercise 8.
A factory produces 1000 components, of which 30 are defective. A quality control inspector randomly selects 10 components for testing. What is the probability that exactly 2 of the selected components are defective?

Solution 8.
In this case, the parameters are:

- $N = 1000$ is the total number of balls (10 red + 15 green),
- $M = 30$ is the number of red balls,
- $n = 10$ is the number of balls drawn,
- $x = 2$ is the number of red balls drawn.

Using the hypergeometric distribution formula:

$$P(X=2) = \frac{\binom{30}{2}\binom{970}{8}}{\binom{1000}{10}}$$

Now, calculate the individual binomial coefficients:

$$\binom{30}{2} = \frac{30 \times 29}{2 \times 1} = 435$$

$$\binom{970}{8} \approx \frac{970 \times 969 \times \cdots \times 963}{8!} \approx 2874890608 \quad \text{(using a calculator)}$$

$$\binom{1000}{10} \approx 2634095600 \quad \text{(using a calculator)}$$

Thus, the probability is approximately:

$$P(X=2) = \frac{435 \times 2874890608}{2634095600} \approx 0.465$$

So, the probability that exactly 2 of the selected components are defective is approximately 0.465.

Exercise 9.
On average, 3 cars pass through a toll booth every minute. What is the probability that exactly 5 cars pass through the toll booth in a minute? (Assume a Poisson distribution.)

Solution 9.
The Poisson distribution is given by the probability mass function:

$$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, \dots,$$

Where:

- $\lambda = 3$ is the average number of cars passing through the toll booth per minute,
- $x = 5$ is the number of cars we want to calculate the probability for.

Now, plug in the values:

$$P(X=5) = \frac{3^5 e^{-3}}{5!}$$

First, calculate the individual components:

$$3^5 = 243$$

$$e^{-3} \approx 0.0498$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Thus, the probability is:

$$P(X=5) = \frac{243 \times 0.0498}{120} \approx 0.100$$

The probability that exactly 5 cars pass through the toll booth in a minute is approximately 0.100.

Exercise 10.
A hospital emergency room receives an average of 2 emergency cases every 30 minutes. What is the probability that exactly 3 emergency cases will arrive in the next 30 minutes?

Solution 10.
Here, we use the Poisson distribution with:

- $\lambda = 3$(average number of emergency cases per 30 minutes),
- $x = 5$ (the number of cases we want to calculate the probability for).

Using the Poisson formula:

$$P(X = 3) = \frac{2^3 e^{-2}}{3!}$$

Now, compute the components:

$$2^3 = 8$$
$$e^{-2} \approx 0.1353$$
$$3! = 3 \times 2 \times 1 = 6$$

Thus, the probability is:

$$P(X = 3) = \frac{8 \times 0.1353}{6} \approx 0.180$$

The probability that exactly 3 emergency cases will arrive in the next 30 minutes is approximately 0.180.